



MECHATRONICS SYSTEM INTEGRATION (MCTA 3203)

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PROGRESS REPORT #2

MICROCONTROLLER-BASED AUTOMATIC WASHING MACHINE WITH LOAD DETECTION

SECTION 2

GROUP 16

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INTRODUCTION

The development of smart home appliances has significantly improved daily efficiency, particularly in laundry management. This project focuses on the design and integration of a Microcontroller-Based Automatic Washing Machine with Load Detection. By utilizing an Arduino UNO as the primary controller, the system automates essential washing cycles while incorporating intelligent sensing to optimize resources.

A key feature of this system is the integration of a load detection mechanism using a 1kg Load Cell and an HX711 Load Cell Amplifier. This allows the machine to determine the weight of the laundry and adjust its operations accordingly. Based on the detected weight, the system provides specific feedback regarding water requirements and operation time:

- Light Loads (< 100g): Operates for 1 hour with 100ml of water.
- Medium Loads (< 200g): Operates for 2 hours with 200ml of water.
- Heavy Loads: Operates for 3 hours with 300ml of water.

The user interface includes a 16x2 I2C LCD for real-time status updates and weight display, alongside Push Buttons for start and stop functions. Visual indicators, such as Green and Red LEDs, are used to signify the machine's current state—whether it is idle, washing, or draining. Through the integration of hardware components and custom firmware, this project aims to create a functional prototype that demonstrates the principles of mechatronics system integration.

HARDWARE COMPONENT INTEGRATION

No	Component	Quantity	Notes
1	Arduino UNO (or compatible)	1	Main controller
2	HX711 Load Cell Amplifier	1	For the weight sensor
3	Load Cell (1kg)	1	Connect to HX711
4	16×2 I2C LCD	1	Displays weight, states, water/time info
5	Push Button (START)	1	Active HIGH
6	Push Button (STOP)	1	Active HIGH
7	Green LED	1	Indicates IDLE / weight display
8	Red LED	1	Indicates WASH/DRAIN
9	Resistors 10kΩ	2	Pull-down for buttons (optional if using internal pull-down)
10	Jumper wires	As needed	For connections
11	Breadboard / PCB	1	Optional, for wiring
12	Power Supply 5V	1	From Arduino or external 5V

Table 1: Components

LCD Pin	Arduino Pin
GND	GND
VCC	5V
SDA	A4
SCL	A5

Table 2: Connection of 16x2 I2C LCD on Arduino.

HX711 Pin	Arduino Pin
VCC	5V
GND	GND
DT / DOUT	D2
SCK / CLK	D3

Table 3: Connection of Load Sensor with HX711 on Arduino.

Button	Arduino Pin	Pull
START	D5	Connect one side to 5V, other to D5 → 10kΩ to GND (pull-down)
STOP	D4	Connect one side to 5V, other to D4 → 10kΩ to GND (pull-down)

Table 4: Connection of Buttons on Arduino.

LED Color	Arduino Pin	Resistor
Green	D6	220 Ω in series to GND
Red	D7	220 Ω in series to GND

Table 5: Connection of LEDs on Arduino.

CODING (ARDUINO)

```
#include "HX711.h"

#include <LiquidCrystal_I2C.h>

const int DOUT_PIN = 2;

const int CLK_PIN  = 3;

const int START_BTN = 5;

const int STOP_BTN  = 4;

const int GREEN_LED = 6;

const int RED_LED   = 7;

LiquidCrystal_I2C lcd(0x27,16,2);

HX711 scale;

float calibration_factor = 998;

float weight = 0.0;

// ===== STATES =====

enum State {IDLE, WASHING, DRAINING, PAUSED};

State currentState = IDLE;

State prevState    = IDLE;

// ===== TIMERS =====
```

```
unsigned long timerStart = 0;

unsigned long remainingTime = 0;

// WASH & DRAIN TIME = 30s

const unsigned long WASH_TIME = 15000;

const unsigned long DRAIN_TIME = 15000;

// ===== BUTTON EDGE =====

bool lastStartBtn = LOW;

bool lastStopBtn = LOW;

// ===== BUTTON LATCH =====

bool stopRequest = false;

void setup() {

    Serial.begin(38400);

    lcd.init();

    lcd.backlight();

    pinMode(START_BTN, INPUT);

    pinMode(STOP_BTN, INPUT);

    pinMode(GREEN_LED, OUTPUT);

    pinMode(RED_LED, OUTPUT);
```

```
scale.begin(DOUT_PIN, CLK_PIN);

scale.set_scale(calibration_factor);


lcd.print("Stabilizing...");

delay(2000);

scale.tare();


lcd.clear();

lcd.print("System Ready");

delay(1000);

lcd.clear();
}

void loop() {

    // ===== READ BUTTONS =====

    bool startNow = digitalRead(START_BTN);

    bool stopNow  = digitalRead(STOP_BTN);

    // LATCH STOP REQUEST

    if(stopNow == HIGH && lastStopBtn == LOW){

        stopRequest = true;

    }


    bool startPressed = (startNow == HIGH && lastStartBtn == LOW);
```

```
lastStartBtn = startNow;

lastStopBtn  = stopNow;


// ===== IDLE =====

if(currentState == IDLE){

    displayWeight();

}


// ===== STOP BUTTON =====

if(stopRequest){

    // First STOP → PAUSE

    if(currentState == WASHING || currentState == DRAINING){

        prevState = currentState;

        remainingTime = getRemainingTime();

        currentState = PAUSED;

        digitalWrite(REDA_LED, LOW);

        digitalWrite(GREEN_LED, LOW);

        lcd.clear();

        lcd.setCursor(0,0);

        lcd.print("      PAUSED      ");

    }

}
```

```
// Second STOP → RESET

else if(currentState == PAUSED){

    currentState = IDLE;

    lcd.clear();

}

stopRequest = false; // consume the request

}

// ===== START BUTTON =====

if(startPressed){

    // Resume after pause

    if(currentState == PAUSED){

        currentState = prevState;

        timerStart = millis();

        lcd.clear();

    }

    // Start from IDLE

    else if(currentState == IDLE){

        lockWeight();

        remainingTime = 0; // reset timer

        currentState = WASHING;
```

```
        timerStart = millis();

    }

}

// ===== WASHING =====

if(currentState == WASHING){

    digitalWrite(REDA_LED, HIGH);

    digitalWrite(GREEN_LED, LOW);

    lcd.setCursor(0,0);

    lcd.print("    WASH    ");

    if(millis() - timerStart >= getTargetTime()){

        currentState = DRAINING;

        remainingTime = 0;           // reset

        timerStart = millis();

        lcd.clear();

    }

}

// ===== DRAINING =====

if(currentState == DRAINING){

    digitalWrite(REDA_LED, HIGH);

    digitalWrite(GREEN_LED, LOW);
```

```

    lcd.setCursor(0,0);

    lcd.print("    DRAIN    ");


    if(millis() - timerStart >= getTargetTime()){

        lcd.clear();

        lcd.setCursor(5,0);

        lcd.print("DONE");

        delay(10000);

        currentState = IDLE;

        lcd.clear();

    }

}

delay(100); // small loop delay
}

// ===== FUNCTIONS =====

// Display live weight
void displayWeight(){

    digitalWrite(RED_LED, LOW);

    digitalWrite(GREEN_LED, HIGH);

    if(scale.is_ready()){

        weight = scale.get_units(5); // fast read

```

```
    if(weight > -2 && weight < 2) weight = 0.0;

    lcd.setCursor(0,0);

    lcd.print("Weight:      ");

    lcd.setCursor(8,0);

    lcd.print(weight,0);

    lcd.print(" g");

}

}

// Lock weight & show water info
void lockWeight(){

    digitalWrite(REDA_LED, HIGH);

    digitalWrite(GREEN_LED, LOW);

    lcd.clear();

    if(weight < 100){

        lcd.print("Water: 100ml");

        lcd.setCursor(0,1);

        lcd.print("Time: 1 hour");

    }

    else if(weight < 200){

        lcd.print("Water: 200ml");

        lcd.setCursor(0,1);

        lcd.print("Time: 2 hour");

    }

}
```

```

    }

    else{

        lcd.print("Water: 300ml");

        lcd.setCursor(0,1);

        lcd.print("Time: 3 hour");

    }

    delay(3000);

    lcd.clear();

}

// Remaining time when paused
unsigned long getRemainingTime(){

    unsigned long elapsed = millis() - timerStart;

    unsigned long target  = getTargetTime();

    if(elapsed >= target) return 0;

    return target - elapsed;

}

// Target time based on state
unsigned long getTargetTime(){

    if(currentState == WASHING || prevState == WASHING)

        return remainingTime == 0 ? WASH_TIME : remainingTime;

```

```
if(currentState == DRAINING || prevState == DRAINING)

    return remainingTime == 0 ? DRAIN_TIME : remainingTime;


return 0;
}
```

Certificate of Originality and Authenticity

This is to certify that we are **responsible** for the work submitted in this report, that **the original work** is our own except as specified in the references and acknowledgment, and that the original work contained herein has not been untaken or done by unspecified sources or persons. We hereby certify that this report has **not been done by only one individual, and all of us have contributed to the report**. The length of contribution to the reports by each individual is noted within this certificate. We also hereby certify that we have **read and understand** the content of the total report, and no further improvement on the reports is needed from any of the individual contributors to the report. We therefore agreed unanimously that this report shall be submitted for **marking**, and this **final printed report** has been **verified by us**.

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